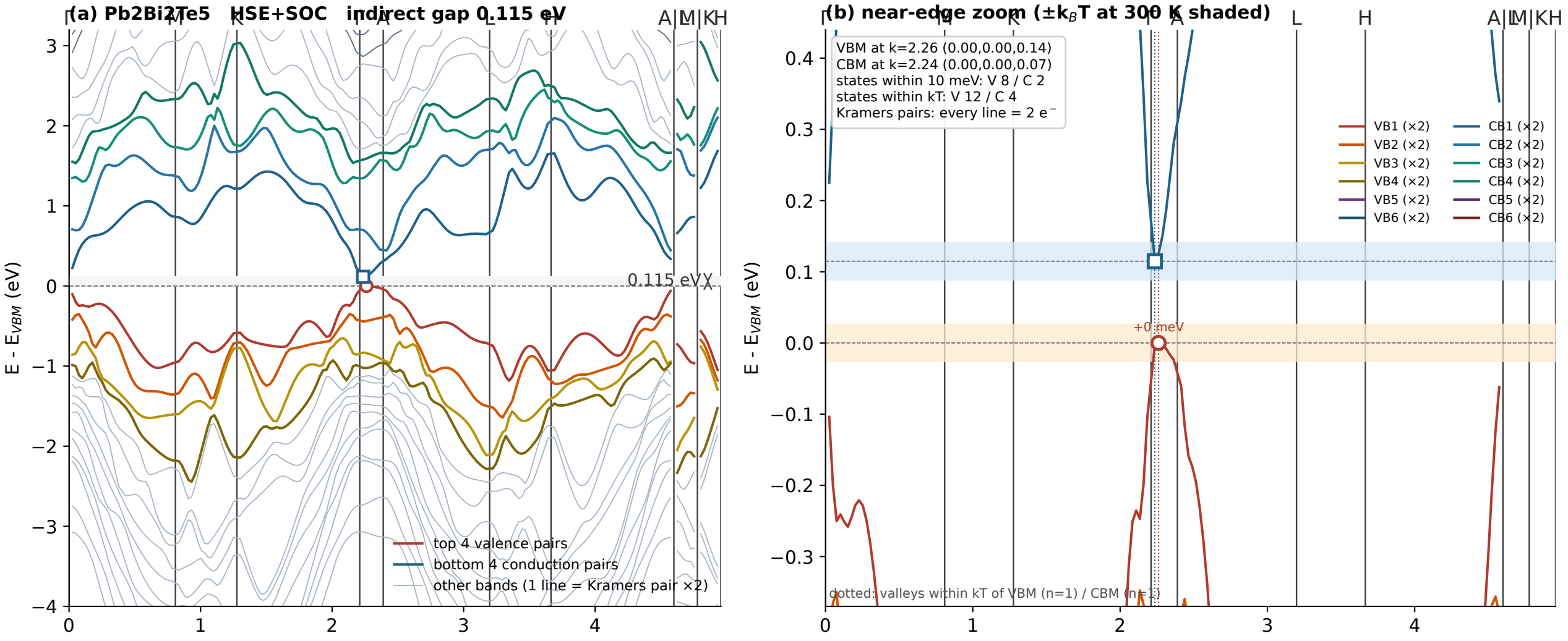

(a) Pb2Bi2Te5 HSE+SOC indirect gap 0.115 eV
Γ M K Γ A L H A|L M|K H
E - E_VBM (eV)
0.115 eV
top 4 valence pairs
bottom 4 conduction pairs
other bands (1 line = Kramers pair ×2)
(b) near-edge zoom (±k_BT at 300 K shaded)
VBM at k=2.26 (0.00,0.00,0.14)
CBM at k=2.24 (0.00,0.00,0.07)
states within 10 meV: V 8 / C 2
states within kT: V 12 / C 4
Kramers pairs: every line = 2 e⁻
VB1 (×2)
VB2 (×2)
VB3 (×2)
VB4 (×2)
VB5 (×2)
VB6 (×2)
CB1 (×2)
CB2 (×2)
CB3 (×2)
CB4 (×2)
CB5 (×2)
CB6 (×2)
+0 meV
dotted: valleys within kT of VBM (n=1) / CBM (n=1)